



DEALER BEST PRACTICE SERIES

Salvage Of Pin Slot Using Standard Sleeve

Maintenance and Repair Application Component Life Management Component Renewal (CRC) MARC Management

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1.0 Introduction

According to Reuse and Salvage guidelines SEBF8257-11, worn or damaged housing pin slots holes can be salvaged by installing sleeves made of SAE 1020 steel. There is no sleeve part number serviced, so dealers typically make their own sleeves with specifications contained in the Reuse and Salvage guide:

1. Bore a hole into the housing in order to accept a sleeve.
 - a. Use a 17.46 mm (0.687 inch) boring tool with a spiral or a straight contour and a 12.7 mm (0.50 inch) tool guide.
 - b. Line up the tool with the unworn portion of the hole.
 - c. Be sure that you only bore the deformed portion of the hole. The depth of the counterbore should allow the sleeve to remain even with the top of the housing surface.
2. Fabricate a sleeve for repair from SAE 1020 steel with a 0.05 mm (0.002 inch) press fit. Drill the sleeve to 12.29 mm (0.484 inch). Ream the sleeve to the correct size after installation.

Ferreyros has identified a supplier of a standard sleeve that meets the specifications required and therefore no longer manufactures our own sleeves. This saves time and reduces cost in the shop.

2.0 Best Practice Description

The fabrication of sleeve in Ferreyros workshop takes about 30 min per sleeve in conventional machine and 10 min per sleeve in CNC machine. The cost of labor as well as the cost of utilizing expensive machining tools was considered

By using a commercially available standard sleeve with measures from ID $\frac{1}{2}$ " x OD $\frac{3}{4}$ " x $\frac{3}{4}$ length, Ferreyros had reduced repair time by 75% and costs approximately 50%. The ID is standard with finish measure from 12.7 +/- 0.005 mm (See the picture 1).

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Picture 1. Dimensions of standard sleeve

The salvage of pin slots with that standard sleeve is used in parts as: 8P3513, 7G5746, 6Y8674, 6Y3297, 3Q6581, 3P2374, 2P4363, 2022730, 1752635, 1109269 and all parts which works with pin slot from ID ½” .

Cost of a standard sleeve is approximately \$15.00 each.

3.0 Implementation Steps

1. Place the sleeves in the freezer for six hours.



Picture 2. Frozen sleeves

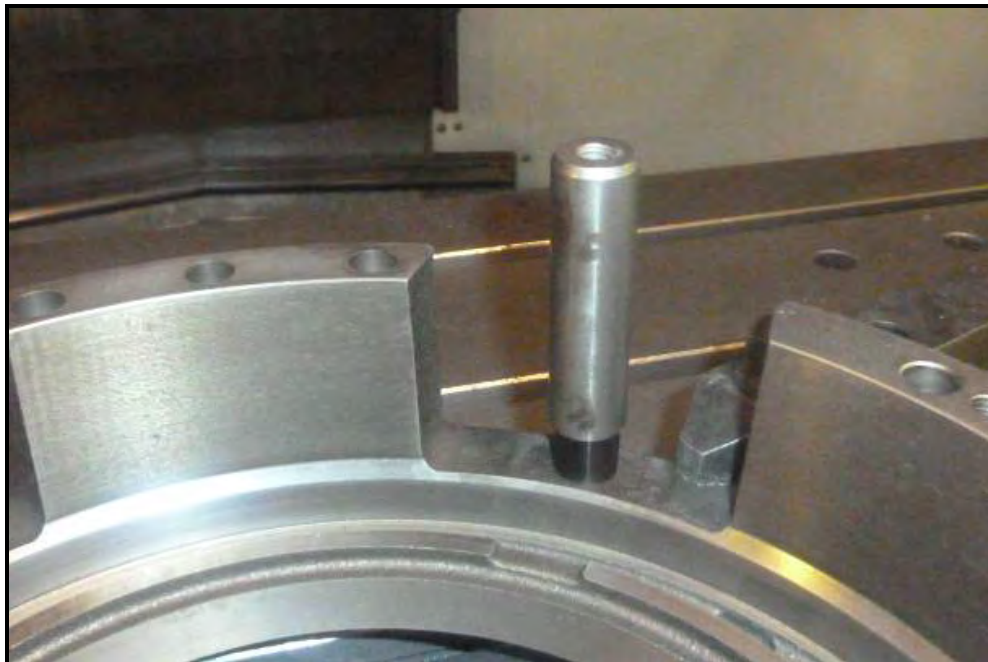
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2. Bore a hole into the housing in order to accept a sleeve. The dimensions should be ID $\varnothing$ 19.07 +/- 0.01 mm x 19.08 +/- 0.02 mm length.



Picture 3. Plate being bored in the HAAS machine

3. Press the standard sleeve into the housing using an installer PIN.



Picture 4. Standard sleeve being installed

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4. The surface of the sleeve should be even with the surface of the housing.



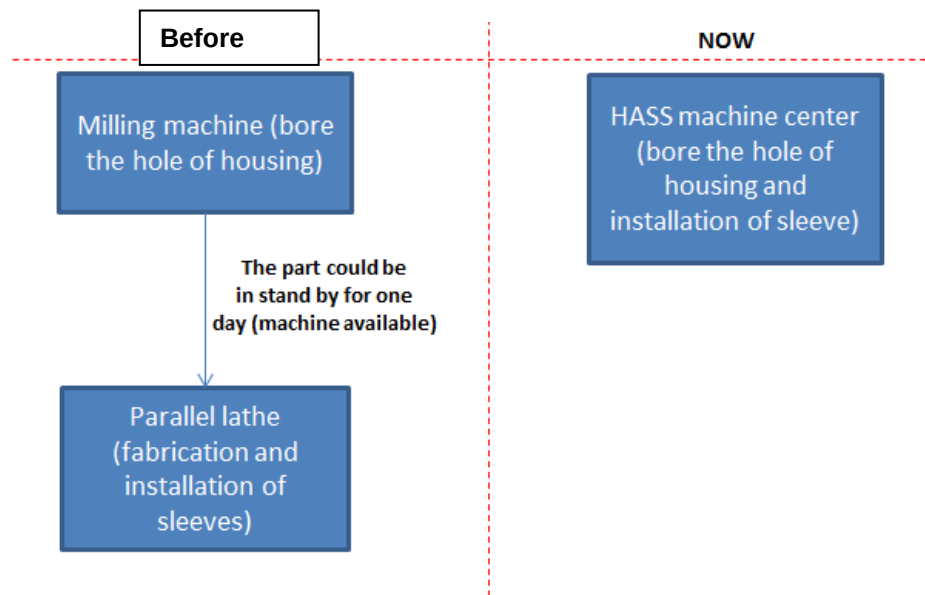
Picture 4. Standard sleeve installed

4.0 Benefit

- Reduce the pin slot time repair by 75%.
- Reduce the use of labor by 80%.
- Reduce the sleeve cost by 50%.
- Use of the Hass machine instead of two machines (See the picture 5).

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Picture 5. Comparison between the use of machines

5.0 Resources Required

EDITOR'S NOTE: Although Caterpillar does not endorse this particular supplier, Ferreyros has experience with the following products or companies. We welcome additional supplier recommendations from our dealer network

- The standard sleeve from ID Ø ½" are provided by MSC Industrial Supply Co.
 - o **Code:** # 09662495 (Gibraltar Products)

6.0 Supporting Attachments / References

None

7.0 Related Best Practices

None

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8.0 Acknowledgements

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